

SkillBridge SG

Cross-Sector Skills Matching for Singapore's Workforce

"Making every transferable skill count"

Alibaba Cloud × Qoder Hackathon Singapore 2026

Live app: <https://skillbridge-sg-1011057703879.asia-southeast1.run.app/>

The Problem

Singapore's SkillsFuture actively encourages **cross-sector career mobility**, yet existing job platforms fail to support it:

- **LinkedIn, MyCareersFuture, JobStreet** use single-sector keyword matching
- A professional accountant with "Data Analytics + Python" is **invisible** to AI Governance roles in ICT — and vice-versa
- Binary skill matching (has / doesn't have) ignores **proficiency levels**
- No lightweight tool understands transferable skills across sectors

Success Metrics

Objective	Demonstration
Ranked results in < 1 second	Live app + latency tests
Proficiency-aware scoring (PL 1 - 6)	matched_detail per result
Cross-sector bridge visible	Bridge chips + confidence
Every recommendation traceable	Data provenance
Inspectable & reproducible	97 - test suite + deterministic seed

What SkillBridge Does

Paste a Job Description → ranked, explained results in **< 1 second**

- Understands **transferable skills** across sectors
- **Proficiency-aware scoring** (PL 1-6, partial credit)
- Every result is **explainable** (matched, gaps, bridges, courses)
- Accessible to recruiters and career coaches without a data-science team

The 8-Stage Pipeline (Core Effort)

```
JD → BM25 (top-50) → Dense (top-50) → RRF Fusion (top-30) →  
Cross-Encoder (top-10) → Explainability → Surprise →  
Narrative → Courses
```

Why hybrid? Lexical (BM25) catches exact keywords but misses semantic similarity. Dense embeddings catch transferable skills but can drift off-topic.

Reciprocal Rank Fusion merges both robustly, and a **cross-encoder** scores each (JD, candidate) pair for precise ranking.

Funnel: 150 candidates → 50+50 recalled → 30 fused → 10 re-ranked → scored & explained

Proficiency-Aware Scoring (Differentiator)

Official SWDA proficiency scale:

1 Use → 2 Operate → 3 Apply → 4 Implement → 5 Analyse → 6 Assess/Lead

Per-skill credit:

- Full credit `1.0` if `candidate_PL ≥ required_PL`
- Partial credit `candidate_PL / required_PL` if below required level
- Zero if skill is absent

Hybrid display score: `40 + 58 × (0.3 × semantic_norm + 0.7 × proficiency_fit)` → range 40-98

70% weight on proficiency fit ensures **real skill match dominates text similarity**

Bridge Skills (Transferable Evidence)

A bridge answers: *"Candidate lacks skill X, but has skill Y — how much of X is covered by Y?"*

Derived from 150K+ Knowledge & Ability (K&A) items in the official SWDA dataset

Containment score: $|S \cap V| / |S|$, floor 0.2, Critical Core Skills (CCS) +0.1 boost

53 evidence-based bridges with explicit confidence scores (0.20-0.87)

Data Provenance (Not a Black Box)

All data grounded in **official SWDA Skills Framework dataset**:

Seed File	Count	Key Details
jobs.json	30 roles	6 × 5 sectors, skill_requirements verbatim
candidates.json	150 profiles	Anchored to real roles, ~78% skill coverage
skill_registry.json	434 skills	TSC/CCS titles, Emerging/CASL flags
skill_taxonomy.json	5 sectors	finance 70, ict 63, healthcare 80, eng 64, sustainability 77
bridges.json	53 bridges	K&A containment, floor 0.2, CCS +0.1 boost
courses.json	434 entries	Live MySkillsFuture search URLs

Deterministic: `random.seed(20260729)` → byte-identical output, auditable, validated

What It Can Do

- ✓ Match JD → ranked results in < 1 second
- ✓ Proficiency-aware scoring (PL 1-6) grounded in official data
- ✓ Cross-sector bridge skills with evidence-backed confidence
- ✓ Explainability panel: matched, gaps, bridges, courses
- ✓ What-if explorer: add/remove skills, see scores shift ▲/▼
- ✓ Hire ↔ Upskill perspective toggle
- ✓ Surprise mode: surface serendipitous candidates
- ✓ Emerging & CASL badges from official dataset
- ✓ Sector convergence strip (Jaccard overlap visualization)
- ✓ 97 tests, fully reproducible, deterministic

Cannot: production-scale talent pool, real resume parsing, open-ended NLP skill extraction

Demo

Live app: <https://skillbridge-sg-1011057703879.asia-southeast1.run.app/>

Video: <https://www.youtube.com/watch?v=DhmpkfX31yg>

GitHub: https://github.com/L3sli3Ch1ang/qoder_hackathon

Live app: FastAPI + Alpine.js + Tailwind

Run locally:

```
# Just cd in – direnv activates the flake shell automatically
cd qoder_cli
uvicorn app.main:app --host 127.0.0.1 --port 8000
```

Built with Qoder

Quest Mode: spec + scaffold, SWDA data integration, 6-phase development

Expert Mode: 8-stage pipeline, proficiency scoring, bridge derivation from 150K+ K&A items

qodercli: test runs, isolated worktree, `/review`, `/init`

2 weeks, 97 tests, real government data, deterministic seed generation

Evidence: `AGENTS.md`, `docs/evidence/`, `docs/evidence/code-reviews/`

Stack: FastAPI + Alpine.js + Tailwind, sentence-transformers, Qdrant Lite, DashScope qwen-plus, NixOS + direnv

Roadmap

Future investments (in priority order):

1. **Real candidate ingestion** — parse PDF/DOCX resumes, map to SWDA taxonomy
2. **Scale corpus** — expand from 5 to 38 SWDA sectors (2,030 roles / 12,007 skills)
3. **Richer skill extraction** — NER/embedding-based detection beyond taxonomy
4. **Live data sync** — periodic re-derivation from SWDA/MySkillsFuture releases
5. **Persistence & accounts** — save searches, track pipelines, multi-user
6. **Outcome feedback loop** — tune blend weights based on actual hires
7. **Bias & fairness review** — audit scorer for sector/seniority bias

Thank You

Questions?

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GitHub: https://github.com/L3sli3Ch1ang/qoder_hackathon/tree/main

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